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C2**Started on** Thursday, 17 September 2026, 8:24 AM**State** Finished**Completed on** Thursday, 17 September 2026, 8:25 AM**Time taken** 38 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  #define MAX_N 100005
4
5  int sequence[MAX_N];
6  int tails[MAX_N];
7
8  // Binary search to find the first element strictly greater than key (upper_bound)
9  int upper_bound(int arr[], int length, int key) {
10     int low = 0, high = length;
11     while (low < high) {
12         int mid = low + (high - low) / 2;
13         if (arr[mid] <= key) {
14             low = mid + 1;
15         } else {
16             high = mid;
17         }
18     }
19     return low;
20 }
21
22 int main() {
23     int n;
24
25     // Read the number of elements
26     if (scanf("%d", &n) != 1) return 0;
27
28     // Read the sequence values
29     for (int i = 0; i < n; i++) {
30         if (scanf("%d", &sequence[i]) != 1) return 0;
31     }
32
33     int len = 0; // Tracks the length of the longest non-decreasing subsequence
34
35     for (int i = 0; i < n; i++) {
36         int idx = upper_bound(tails, len, sequence[i]);
37
38         tails[idx] = sequence[i];
39
40         if (idx == len) {
41             len++; // Expanded the length of the subsequence
42         }
43     }
44
45     // Output the maximum length found
46     printf("%d\n", len);
47
48     return 0;
49 }
50

```

	Input	Expected	Got	
✓	9 -1 3 4 5 2 2 2 2 3	6	6	✓
✓	7 1 2 2 4 5 7 6	6	6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)